

Program : Diploma in Civil and Environmental Engineering	
Course Code : 3357	Course Title: Building Materials and Construction
Semester : 3	Credits: 4
Course Category: Programme core Course	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge about availability, manufacturing, processing, suitability and characteristics of materials for various applications in construction.
- To familiarize the various components of buildings and their constructional procedures.

Course Prerequisites:

Topic	Course code	Course name	Semester

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Understand ingredients of concrete, manufacturing of concrete and properties of fresh and hardened concrete and suggest suitable type of cement, admixture, and manufacturing method based on field conditions.	18	Applying
CO2	Understand characteristics of various types of natural, artificial and processed construction materials and select relevant types of construction materials for site conditions in tune with sustainability.	12	Applying

CO3	Understand types of structures, construction of components of substructure and suggest suitable type of foundation based on site conditions.	12	Applying
CO4	Understand types and construction methods of various components of superstructure and suggest suitable type of each component based site conditions and requirements.	16	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1							
CO2							
CO3							
CO4							

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand ingredients of concrete, manufacturing of concrete, and properties of fresh and hardened concrete and suggest suitable type of cement, admixture and manufacturing methods based on field conditions.		
M1.01	Understand role of cement in concrete, manufacturing of cement, hydration of cement, tests on cement and suggest suitable type of cement based requirements.	5	Applying
M1.02	Understand role of aggregate in concrete and types of coarse and fine aggregate.	4	Understanding
M1.03	Understand role of admixtures in concrete and types of chemical admixtures and suggest suitable types based on requirements.	1	Applying
M1.04	Understand quality requirements of water for concreting	1	Understanding
M1.05	Understand various stages of manufacturing of concrete and suggest methods to be adopted for transportation, placing, curing and compaction of	3	Applying

	concrete based on site conditions		
M1.06	Understand properties of fresh and hardened concrete.	4	Understanding
Contents: Concrete- Introduction – Scope and significance of concrete in construction industry. Ingredients, Role of each ingredient in concrete. Cement- Ordinary Portland cement- manufacturing (flow chart of wet and dry process), hydration of cement, setting and hardening of cement, special cement- Portland pozzolanic cement, Portland slag cement, rapid hardening cement, quick setting cement, sulphate resisting cement, low heat cement, geopolymer cement (properties, application, manufacturing, and eco-friendly aspect). Brief description on tests to be conducted on cement- field tests and lab tests. IS specifications for OPC regarding fineness, setting time, soundness, grade. Aggregate- Classification of aggregate. Role of coarse and fine aggregate in concrete. - Requirements of good coarse aggregate, Classification of coarse aggregate based on shape and size. Fine aggregate- Zones, Classification based on source. Tests on aggregate- sieve analysis, crushing strength, impact value, abrasion test (brief description only). Grading requirement. Chemical admixtures- types and uses. Water quality for concreting. Concrete- Mix design- Nominal mix and design mix- water cement ratio. Brief description on stages of manufacturing- Batching, mixing, transportation, placing- formworks used, compaction, curing. Properties of fresh concrete- Workability, segregation, bleeding. Slump test, compaction factor test and vee bee consistometer test for workability. Hardened concrete- Grade of concrete, Test to find out compressive strength of concrete. Factors affecting durability of concrete.			
CO2	Understand characteristics of various types of natural, artificial, and processed construction materials and select relevant type of construction materials in tune with sustainability.		
M2.01	Understand characteristics of good building stones and types of commonly used stones for construction purpose	1	Understanding
M2.02	Understand methods adopted for seasoning and preservation of timber and bamboo.	1	Understanding
M2.03	Understand properties of various wood products available in market for construction purpose and suggest suitable type based on requirements.	2	Applying
M2.04	Understand types of bricks available in market	3	Applying

	and suggest suitable type based on requirements		
M2.05	Understand uses and types of metals, glass, paints, water proofing and thermal proofing materials for construction.	3	Understanding
M2.06	Understand scope of use of agricultural waste products, industrial by product- Plaster of Paris as construction material.	2	Understanding
	Series Test -I	1	

Contents: Building stones- Requirements of good building stones, Commonly used building stones in India for flooring and masonry works.

Timber- Properties of good timber, seasoning and preservation of timber, use of bamboo in construction. Wood products: Plywood, particle board, veneers, laminated board, MDF, HDF, WPC –uses and suitability in different construction works/ site conditions.

Brick- Types of brick- clay brick, cement concrete bricks, fly ash bricks, Autoclaved Aerated blocks (brief description on properties, uses, available sizes, physical requirements as per IS specifications if any and suitability for various site conditions.). Field tests on bricks.

Glass- Uses, properties of soda lime glass, lead glass, toughened glass and borosilicate glass. Suitability of glass/ type of glass in different construction works.

Ferrous and non ferrous metals-iron, steel, lead, copper (properties and uses), Aluminium-properties, market forms, ACP.

Paints- Ingredients. Types- whitewash, cement paint, distempers, oil paints and varnishes (only properties and suitability of each type for finishing works.).

Waterproofing and thermal proofing materials.

Constituents and uses of Plaster Of Paris, POP finishing boards, sizes and uses.

Use of agricultural waste – rice husk, bagasse, coir fibres in construction materials.

CO3	Understand types of structures, construction of components of substructure and suggest suitable type of foundation based on site conditions.		
M3.01	Understand classification of buildings based on relevant codes and types of structures.	3	Understanding
M3.02	Understand components of substructure and superstructure of a building	1	Understanding
M3.03	Understand construction stages of substructure	3	Understanding
M3.04	Understand different types of foundation and suggest a type of foundation based on site conditions.	4	Applying

M3.05	Understand need and methods of underpinning	1	Understanding
Contents: Classification of buildings as per National Building Code/ KMBR/KPBR. Types of construction- Load bearing wall structure, framed structure, composite structure, prefabricated structure, special framed structure for earthquake resistance (IS 13920). Construction methods for structural steel buildings. Building components- functions of building components- substructure- foundation-plinth-DPC. Super structure- Parts of building- Walls, partition wall and cavity wall, sill, Lintel, doors and windows, floor etc. Construction of substructure- Job layout- Site clearance, Layout for load bearing structure and framed structure by center line and face line method. Earthwork: Excavation for foundation- Timbering and strutting- Material used for plinth filling- Foundation- Functions of foundation- Types of foundation- shallow foundation and deep foundation- stepped footing, wall footing, column footing, isolated and combined column footing, raft foundation, grillage foundation, strap footing, pile foundation, well foundation and Caissons- Suitability of each type for site conditions. Underpinning- need and methods.			
CO4	Understand types and construction methods of various components of superstructure and suggest suitable type of each component based on site conditions and requirements.		
M4.01	Understand types, requirements and suitability of stone masonry and brick masonry and suggest suitable type based on site conditions and requirements.	4	Applying
M4.02	Understand purpose and types of scaffolding and shoring.	2	Understanding
M4.03	Understand types of doors and windows and suggest suitable type based on requirement.	3	Applying
M4.04	Understand vertical communication methods adopted in buildings.	3	Understanding
M4.05	Understand types of flooring and roofing materials adopted and suggest suitable type based on requirements and site conditions.	4	Applying
	Series Test -II	1	
Contents: Types of stone masonry- Rubble masonry, Ashlar masonry. Pointing of stone masonry and their purpose- precautions. Situations in which stone masonry is adopted. Brick masonry- Terms used brick masonry- bonds in brick masonry- header bond, stretcher bond, English bond, Flemish bond etc. Requirements of brick masonry- precautions to be			

observed- hollow and solid concrete block masonry- composite masonry- interlock bricks.
Scaffolding and shoring- purpose and types
Doors- Fully paneled doors, partly paneled and glazed doors, flush doors, collapsible doors, rolling shutters, revolving doors, glazed doors.
Windows: Fully paneled, partly paneled and glazed, sliding windows, louvered window, bay window, corner window, gable and dormer window, skylight- sizes of window recommended by BIS.
Vertical communication methods- stair, lifts and escalators.
Roofs- Roofing materials- RCC, MP tiles, ceramic roofing tiles, Thatched, G I sheets, Corrugated GI sheets, Plastic and fiber sheets, innovative roofing materials (Fiber reinforced cement board, shingles etc). Types of roof- Flat roof, pitched roof – terms used in roofs- King post truss, queen post truss.
Flooring materials- Clay tiles, vitrified tiles, stone tiles, wooden flooring, cement flooring.

Text / Reference:

T/R	Book Title/Author
T1	Ghose D. N, Construction Materials, Tata McGraw Hill, New Delhi
T2	Concrete Technology, Gambhir M L, Tata McGraw Hill, New Delhi
T3	Rangwala S.C, Building Construction, Charotar Publication
R1	Concrete Technology, Shetty M.S, Chand and Co. Pvt, Ltd., Ram Nagar, Delhi
R2	Concrete Technology, Santhakumar A. R, Oxford University Press, New Delhi
R3	Concrete Technology, Neville, A.M, Pearson Education Pvt, Ltd, New Delhi
R4	Rangwala S. C, Engineering Materials, Charotar Publication
R5	Sushil Kumar, Building Construction, Standard Publication

Online resources:

Sl.No.	Website Link
1	NPTEL: Civil Engineering- NOC: Basic construction materials. https://archive.nptel.ac.in/courses/105/106/105106206/
2	NPTEL: Civil Engineering- Civil Engineering- Building materials https://archive.nptel.ac.in/courses/105/102/105102088/
3	https://ekumbh.aicte-india.org